Strength in Numbers: A/B Testing of Swarm Fitness Evaluation

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Evolutionary Robotics Final Project

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Contents

1 Goals

This project extends my four-legged robot deliverables into a swarm context and investigates two fitness evaluation methods.

- **Deliverables recap**:
  1. Multi-URDF output for swarm members (Milestone 1)
  2. Swarm simulation with obstacles (Milestones 2 and 3)
  3. A/B testing framework comparing max-based vs avg-based selection (Milestone 4)
- **A vs B variants**:
  - A: max(distance) across swarm drives selection.
  - B: average(distance) across swarm drives selection.
- **Research question**: Under what conditions does selecting the best individual outperform using the swarm's average distance?

2 Implementation

I built on the existing repo to enable swarm-based evaluation and automate the A/B tests.

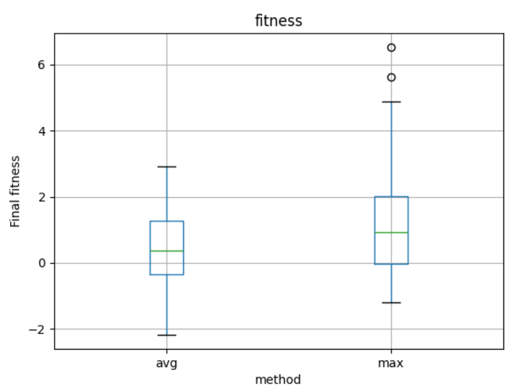
- **Body generation (solution.py)**: wrote out multiple `body_*.urdf` files with per-robot position offsets.
- **Simulation harness (simulate.py & robot.py)**: loaded all URDFs in `src/data/`, ran PyBullet steps, and collected each robot's x-position each timestep.
- **Fitness writer (robot.Get_Fitness)**: computed both max and average distances, wrote `fitness_max<ID>.txt` and `fitness_avg<ID>.txt`, then renamed the chosen metric based on the `FITNESS_METHOD` environment variable.
- **Automation scripts**:
  - `run_ab.sh` to launch 50 replicates per method and gather `fitness_ab_test.csv`.
  - `abtest.py` for statistical analysis (two-sample t-tests) and box-plot generation.

3 Results

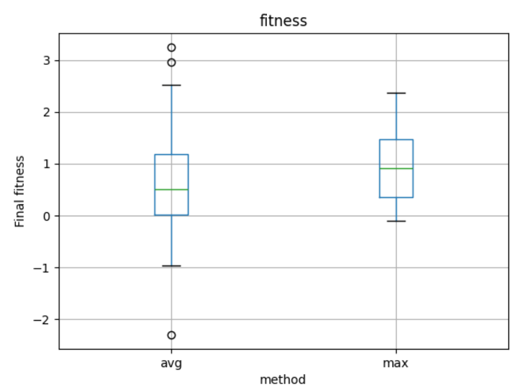
I ran nine experimental configurations (Table 1) and evaluated significance at $\alpha = 0.05$.

Table 1: Summary of A/B test results across experiments

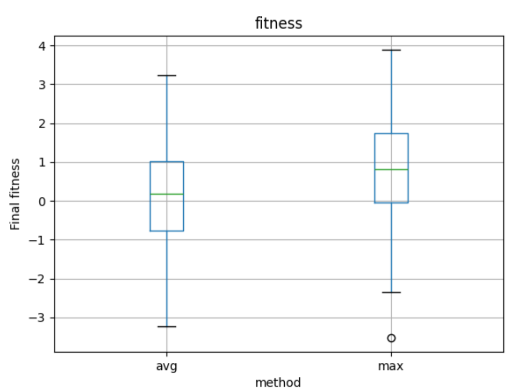
Exp	Obstacles?	Reps	Steps	Pop	Swarm	Max mean	Avg mean	p-value
a	Yes	50	2500	2	2	1.269	0.495	0.0130
b	Yes	50	5000	4	2	1.549	0.940	0.1975
c	Yes	50	1000	2	2	0.942	0.593	0.0377
d	No	50	2500	5	5	0.802	0.174	0.0343
e	No	50	1000	2	2	0.225	-0.036	0.1929
f	Yes	25	1000	2	1	-0.096	0.399	0.0512
g	Yes	50	2000	3	3	1.158	0.243	0.0009
h	No	50	2000	2	3	0.748	0.269	0.1220
i	Yes	50	3000	4	4	0.908	0.532	0.1760



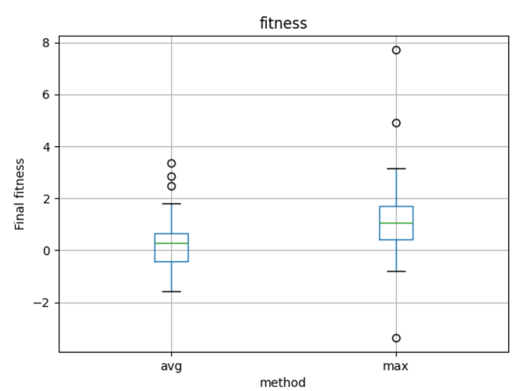
(a) Experiment a



(b) Experiment c



(c) Experiment d



(d) Experiment g

Figure 1: Key experiments where max-based selection was significant (a, c, d, g).

4 Discussion & Future Work

Ease and difficulty: Adapting the fitness writer and automation scripts was straightforward; handling asynchronous simulation outputs and cleanup required careful file-watching logic.

Patterns and Conclusions:

1. Obstacles + moderate size → “max” wins decisively (a, c, g).
2. No obstacles → “max” rarely helps (d marginally; e, h non-significant).
3. High budgets (lots of steps or generations) or large swarms/populations → the advantage of rewarding only the best performer fades (b, i).
4. Single-robot “swarms” collapse to no difference (f).

Future directions:

- Introduce dynamic obstacles, varied terrain, or moving hazards to stress-test swarm coordination.
- Enhance Pyrosim with built-in swarm primitives.
- Scale to larger swarm sizes.